THE PICARD NUMBER OF FIBRED MORI DREAM SURFACES

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ABSTRACT. Let S be a smooth projective surface endowed with a fibration $f : S \longrightarrow C$ onto a smooth projective curve. We prove that, if the pseudo-effective cone $\overline{\mathrm{Eff}}(S)$ is polyhedral, then the Picard number $\rho(S)$ is determined by the number of irreducible components of the reducible fibres of f.

In the special case where S is a Mori dream space, our argument provides a new proof, based on different methods, of the corresponding equality, which also follows from a result of Casagrande.

As a further consequence, if $f : S \longrightarrow \mathbb{P}^1$ is an elliptic fibration with a section and S is a Mori dream space, then the Mordell–Weil group of its generic fibre is finite.

The main results of the paper are applications of this non-polyhedrality criterion. We prove that certain Horikawa surfaces, Fermat surfaces in $\mathbb{P}^3$ of every degree $d \geq 4$, some product-quotient surfaces, and the minimal simply connected numerical Godeaux surface constructed by Craighero and Gattazzo have non-polyhedral pseudo-effective cones. In particular, none of these surfaces is a Mori dream space.

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1. Introduction

Since the pioneering work of Hu and Keel [29], Mori dream spaces have been intensively studied. In particular, considerable attention has been devoted to determining which algebraic varieties are Mori dream spaces and which are not.

Research on this topic has so far concerned mainly, though not exclusively, rational and Fano varieties. By contrast, varieties of general type remain much less understood, and comparatively few examples have been investigated; see, for instance, [11, 26, 33]. The present work contributes to this direction.

The key result underlying our applications is the following, see Theorem 3.4:

Theorem A. *Let $f : S \longrightarrow C$ be a fibration from a smooth complex projective surface S onto a smooth projective curve C. If the pseudo-effective cone $\overline{\mathrm{Eff}}(S)$ is polyhedral, then*

$$\rho(S) = n(f) + 2.$$

Here $\rho(S)$ denotes the Picard number of S. The integer $n(f)$ is a weighted count of the reducible fibres: a fibre with exactly k distinct irreducible components contributes $k-1$ to $n(f)$, see Definition 3.1.

When S is a Mori dream space, the equality in Theorem A is also a special case of a result by Casagrande [10, Corollary 2.16], applied to the fibration $f : S \longrightarrow C$. Indeed, such a fibration is a special contraction, $\rho(C) = 1$, the image in $N_1(S)$ of the numerical group of a general fibre has dimension one, and the remaining terms in Casagrande's formula are precisely the contributions $k_p(f) - 1$ of the reducible fibres. Our proof uses different methods and, more importantly, requires only the weaker assumption that $\overline{\mathrm{Eff}}(S)$ is polyhedral.

In particular, Theorem A implies that, if $\overline{\mathrm{Eff}}(S)$ is polyhedral, then all fibrations on S have the same weighted number $n(f)$ of reducible fibres. This is not true in general: for example, we exhibit two fibrations on the Fermat quartic surface with different values of $n(f)$, see Remark 6.6.

Taking the contrapositive of Theorem A, we obtain the following numerical criterion for non-polyhedrality. Since the pseudo-effective cone of a Mori dream space is polyhedral, the criterion also provides an obstruction to the Mori dream property, see Corollary 3.5.

Corollary B. *Let S be a smooth projective surface endowed with a fibration $f : S \longrightarrow C$. If*

$$\rho(S) > n(f) + 2,$$

then $\overline{\mathrm{Eff}}(S)$ is non-polyhedral. In particular, S is not a Mori dream space.

Although this article focuses mainly on surfaces of general type, our result can also be useful in other contexts. For instance, it yields a finiteness result for the Mordell–Weil group of the generic fibre of an elliptic fibration, see Corollary 3.8.

We apply this circle of ideas to several surfaces and families of surfaces of general type with $q = 0$, namely:

- certain double covers of Hirzebruch surfaces, including Horikawa surfaces for every integer $p_g \geq 3$, that is, minimal surfaces of general type satisfying $K^2 = 2p_g - 4$ (Theorem 5.4 and Corollary 5.5);
- Fermat surfaces in $\mathbb{P}^3$ of every degree $d \geq 4$ (Theorem 6.5);
- certain product-quotient surfaces, namely minimal resolutions of quotients $(C_1 \times C_2)/G$, where a finite group G acts diagonally on a product of curves (Theorems 7.7, 7.11, and 7.14);
- the Craighero-Gattazzo surface, a simply connected minimal surface of general type with $p_g = 0$ and $K^2 = 1$ (Theorem 8.2).

In each case, we compare the Picard number with the weighted number of reducible fibres of a suitable fibration and apply Corollary B. We thereby prove that the pseudo-effective cone is non-polyhedral and, consequently, that the surface is not a Mori dream space.

Our work was motivated by a question posed by Keum and Lee [33], who asked whether there exists a minimal surface of general type with $p_g = q = 0$ that is not a Mori dream space. We show that the Craighero-Gattazzo surface provides an affirmative answer to this question. Other examples were recently constructed in [11]; all of them have Picard number 2, and therefore have polyhedral pseudo-effective cones. By contrast, the pseudo-effective cone of the Craighero-Gattazzo surface is non-polyhedral, so our example is of a substantially different nature.

We note that the assumption $p_g = 0$ in [33] is intended mainly to simplify the computation of the Picard number. The examples with $p_g > 0$ constructed in this paper are therefore also relevant to the broader problem.

We conclude the introduction by describing the organization of the paper. Section 2 recalls some preliminary results on Mori dream surfaces. Section 3 contains the proof of Theorem A and Corollary B, together with the application to elliptic fibrations. In the subsequent sections, we prove that several families of surfaces of general type have non-polyhedral pseudo-effective cones by studying the reducible fibres of suitable fibrations.

Notation and conventions We work over the field $\mathbb{C}$ of complex numbers.

Let X be a normal, $\mathbb{Q}$-factorial projective variety, and let K_X be a canonical Weil divisor. We define the geometric genus and the irregularity of X by

$$p_g(X) := h^0(X, \mathcal{O}_X(K_X)) = h^0(X, \omega_X), \qquad q(X) := h^1(X, \mathcal{O}_X).$$

We denote numerical equivalence of divisors by $\equiv$.

A *fibration* is a surjective morphism $f : X \longrightarrow Y$ with connected fibres between normal projective varieties such that $\dim Y < \dim X$.

2. Preliminaries: Mori dream surfaces

If X is a normal $\mathbb{Q}$-factorial projective variety, we set

$$N^1(X)_{\mathbb{Q}} := (\mathrm{Pic}(X)/\equiv) \otimes_{\mathbb{Z}} \mathbb{Q}, \qquad \text{and} \qquad N^1(X) := \big(\mathrm{Pic}(X)/\equiv\big) \otimes_{\mathbb{Z}} \mathbb{R}.$$

The dimension of $N^1(X)$ as a real vector space is the *Picard number* $\rho(X)$ of X. We denote by $N_1(X)$ the real vector space of 1-cycles modulo numerical equivalence.

Since X is $\mathbb{Q}$-factorial, every Weil divisor determines a class in $N^1(X)$. The *effective cone* of X is the convex cone generated by the numerical classes of effective Weil divisors:

$$\mathrm{Eff}(X) := \left\{ \sum_{i=1}^{r} a_i [D_i] \;\middle|\; a_i \in \mathbb{R}_{\geq 0},\ D_i \text{ is an effective Weil divisor} \right\} \subset N^1(X).$$

The *pseudo-effective cone* $\overline{\mathrm{Eff}}(X)$ is the closure of $\mathrm{Eff}(X)$ in $N^1(X)$.

The *Mori cone* $\overline{\mathrm{NE}}(X) \subset N_1(X)$ is the closure of the cone generated by numerical classes of effective curves. If $\dim X = 2$, the identification of Weil divisors with 1-cycles induces a natural isomorphism

$$N^1(X) \simeq N_1(X),$$

under which $\overline{\mathrm{Eff}}(X)$ is identified with $\overline{\mathrm{NE}}(X)$.

The *nef cone* $\mathrm{Nef}(X) \subseteq N^1(X)$ is the cone consisting of the numerical classes of nef divisors.

A $\mathbb{Q}$-Cartier divisor D on X is called *semi-ample* if some positive integral multiple of D is Cartier and its complete linear system is base-point-free. The *semi-ample cone* $\mathrm{SAmp}(X)$ is the cone generated by the numerical classes of semi-ample divisors.

A cone in $N^1(X)$ is called *polyhedral* if it is the positive hull of finitely many vectors. In particular, every polyhedral cone is closed.

For the following characterization of Mori dream spaces, see [29, Definition 1.10 and Proposition 2.9].

Definition 2.1. A normal projective variety X is a *Mori dream space* if X is $\mathbb{Q}$-factorial, the equality

$$\mathrm{Pic}(X)_{\mathbb{Q}} = N^1(X)_{\mathbb{Q}},$$

holds and the Cox ring of X is finitely generated.

For a smooth complex projective variety X, the condition $\mathrm{Pic}(X)_{\mathbb{Q}} = N^1(X)_{\mathbb{Q}}$ is equivalent to $q(X) = 0$. Thus, for the smooth projective surfaces considered in this paper, Definition 2.1 amounts to requiring that $q(S) = 0$ and that the Cox ring of S be finitely generated.

For every smooth projective surface S, we have the inclusions

$$\mathrm{SAmp}(S) \subseteq \mathrm{Nef}(S) \subseteq \overline{\mathrm{Eff}}(S),$$

see [2, Proposition 1.2]. Moreover, with respect to the intersection pairing, $\mathrm{Nef}(S)$ is the dual cone of $\overline{\mathrm{Eff}}(S)$. Consequently, if $\overline{\mathrm{Eff}}(S)$ is polyhedral, then $\mathrm{Nef}(S)$ is polyhedral as well. In particular, if $\mathrm{Eff}(S)$ is polyhedral, then it is closed and hence $\mathrm{Eff}(S) = \overline{\mathrm{Eff}}(S)$, so $\mathrm{Nef}(S)$ is also polyhedral.

The following result is [1, Corollary 2.6].

Theorem 2.2. *Let S be a smooth projective surface with $q(S) = 0$. Then S is a Mori dream space if and only if $\mathrm{Eff}(S)$ is polyhedral and $\mathrm{Nef}(S) = \mathrm{SAmp}(S)$.*

3. The main theorem

We begin by defining the integer $n(f)$, which measures the number of reducible fibres of f, counted with a suitable weight.

Definition 3.1. Let $f : S \longrightarrow C$ be a fibration from a smooth projective surface S onto a smooth projective curve C. For each $p \in C$, let $k_p(f)$ denote the number of irreducible components of the set-theoretic fibre $f^{-1}(p)$, and set

$$n(f) := \sum_{p \in C} \big(k_p(f) - 1\big). \tag{3.1}$$

By generic smoothness, all but finitely many fibres of f are smooth. Since the fibres of f are connected, every smooth fibre is irreducible; hence the sum in (3.1) is finite. Notice also that an irreducible fibre contributes zero to $n(f)$, even if it is non-reduced. Thus the non-zero terms in (3.1) correspond precisely to the reducible fibres of f; in particular, $n(f) = 0$ if and only if every fibre of f is irreducible.

We first prove one inequality in Theorem A, namely $\rho(S) \geq n(f)+2$; this inequality holds without any assumption on $\overline{\mathrm{Eff}}(S)$.

Let F be a smooth fibre of f, and denote its numerical class in $N^1(S)$ by $e = [F]$. We also denote by $e_1, \ldots, e_m$ the numerical classes of all the irreducible components of the reducible fibres of f, where $m = 0$ if f has no reducible fibres.

Lemma 3.2. *With the notation above, we have*

$$\dim_{\mathbb{R}} \mathrm{span}(e, e_1, \ldots, e_m) = n(f) + 1.$$

Consequently, we get

$$\rho(S) \geq n(f) + 2. \tag{3.2}$$

Proof. Let us show how to extract from the set $\{e, e_1, \ldots, e_m\}$ a basis of the corresponding vector subspace. Each reducible fibre $f^{-1}(p)$ of $f : S \longrightarrow C$ gives a linear relation in $N^1(S)$ involving e and the classes of components of $f^{-1}(p)$. If r is the number of reducible fibres of f, we have $r = m - n(f)$ linear relations and so we are left with

$$m + 1 - r = m + 1 - (m - n(f)) = n(f) + 1$$

generators, which are linearly independent by the classical lemma of Zariski [3, Chap. III, Lemma 8.2].

Finally, every class in $\mathrm{span}(e, e_1, \ldots, e_m)$ is orthogonal to e, whereas $he > 0$ for every ample class h. Hence this span is a proper subspace of $N^1(S)$, and (3.2) follows. □

In order to prove the reverse inequality we need the following variant of a result by Rabindranath [46, Proposition 2.1], for which we give here a slightly different proof.

Proposition 3.3. *Let S be a smooth projective surface. Suppose that one can find three non-zero classes $e, v, h \in N^1(S)$, with e pseudo-effective and h ample, such that*

$$e^2 = ev = hv = 0$$

and moreover

$$(e + \mathbb{R}v) \cap \overline{\mathrm{Eff}}(S) = \{e\}. \tag{3.3}$$

Then $\overline{\mathrm{Eff}}(S)$ is non-polyhedral.

Proof. Since $e^2 = 0$ and $h^2 > 0$, the classes e, h generate a 2-dimensional subspace $\mathrm{span}(e, h) \subset N^1(S)$. Since Kleiman's criterion [34] implies $he > 0$, a straightforward calculation shows that the radical of the restriction of the intersection form to $\mathrm{span}(e, h)$ is trivial. But $v \in \mathrm{span}(e, h)^\perp$, so e, v, h are linearly independent and the subspace $W := \mathrm{span}(e, v, h) \subset N^1(S)$ has dimension 3. By the Hodge Index theorem the restriction of the intersection form to $h^\perp$ is negative definite: then, as $\dim(W \cap h^\perp) = 2$, the intersection form on W has signature $(1, 2)$.

After rescaling h, we may assume that $h^2 = 1$, and we can choose an orthogonal basis $\{h, w_1, w_2\}$ of W such that $w_1^2 = w_2^2 = -1$. With respect to this basis, setting $w = xh + y_1 w_1 + y_2 w_2$, the isotropic cone $\{w^2 = 0\} \subset W$ has equation $x^2 - y_1^2 - y_2^2 = 0$. Thus, the intersection of this cone with the affine plane

$$H := \{w \in W \mid (w - e)h = 0\},$$

which is orthogonal to the x-axis and does not contain the origin, is the circle C of equation

$$y_1^2 + y_2^2 = (he)^2.$$

Let U be the bounded open region enclosed by C; then every $w \in U$ satisfies

$$w^2 > 0 \quad \text{and} \quad hw > 0. \tag{3.4}$$

Choose a sequence of rational classes $w_i \in N^1(S)_{\mathbb{Q}}$ converging to w. Since both inequalities in (3.4) are open conditions, for sufficiently large i we have

$$w_i^2 > 0 \quad \text{and} \quad hw_i > 0.$$

By the Riemann-Roch theorem [27, Chapter V, Corollary 1.8], a sufficiently divisible integral multiple of a divisor representing w_i is linearly equivalent to an

effective divisor. Therefore $w_i \in \mathrm{Eff}(S)$ and so, passing to the limit, $w \in \overline{\mathrm{Eff}}(S)$. It follows that

$$\overline{U} \subseteq \overline{\mathrm{Eff}}(S).$$

Finally, if $\overline{\mathrm{Eff}}(S)$ were polyhedral, then

$$P := H \cap \overline{\mathrm{Eff}}(S)$$

would be a (possibly unbounded) convex polygonal region containing $\overline{U}$. By (3.3), the affine line $e + \mathbb{R}v$, which is contained in H, intersects P only at e. Thus, the point e would be a vertex of P. However, $e \in C = \partial U$, and a convex polygonal region containing the closed disk $\overline{U}$ cannot have a vertex on the smooth boundary C. Therefore $\overline{\mathrm{Eff}}(S)$ is non-polyhedral. □

We can complete the proof of Theorem A by showing the inequality $\rho(S) \leq n(f) + 2$. Our argument refines the proof of a weaker inequality due to Rabindranath [45, Theorem 12, p. 36].

Theorem 3.4. *Let $f : S \longrightarrow C$ be a fibration of a smooth complex projective surface S onto a smooth projective curve C. If $\overline{\mathrm{Eff}}(S)$ is polyhedral, then equality in (3.2) holds, namely*

$$\rho(S) = n(f) + 2.$$

Proof. It is enough to prove that, if $\rho(S) > n(f) + 2$, then $\overline{\mathrm{Eff}}(S)$ is non-polyhedral (see Lemma 3.2).

Consider the pushforward map $f_* : N^1(S) \simeq N_1(S) \longrightarrow N_1(C) \simeq \mathbb{R}$. An ample class intersects the fibre e positively, so f_* is non-zero and $\ker f_* \subset N^1(S)$ has codimension 1.

Any irreducible component of a fibre is contracted by f, so we have the inclusion

$$\mathrm{span}(e, e_1, \ldots, e_m) \subseteq \ker f_*. \tag{3.5}$$

By Lemma 3.2, if $\rho(S) > n(f) + 2$ then the inclusion in (3.5) is strict, because

$$\dim \ker f_* = \rho(S) - 1 > n(f) + 1 = \dim_{\mathbb{R}} \mathrm{span}(e, e_1, \ldots, e_m).$$

This means that we can find $v \in \ker f_*$ such that

$$v \notin \mathrm{span}(e, e_1, \ldots, e_m). \tag{3.6}$$

Since $v \in \ker f_*$, we have $ev = 0$. Let us fix an ample divisor h; then, up to adding to v a suitable multiple of e, we may assume $hv = 0$.

Let us consider now the affine line $e + \mathbb{R}v \subset \ker f_*$. A result of Debarre, Jiang, and Voisin [20, Theorem 5.7] implies that

$$\overline{\mathrm{Eff}}(S) \cap \ker f_* = \mathbb{R}_{\geq 0} \cdot e + \mathbb{R}_{\geq 0} \cdot e_1 + \ldots + \mathbb{R}_{\geq 0} \cdot e_m,$$

and so, because of (3.6), we have

$$(e + \mathbb{R}v) \cap \overline{\mathrm{Eff}}(S) = \{e\}.$$

Summing up, the three classes $e, v, h \in N^1(S)$ satisfy all the assumptions in Proposition 3.3, hence $\overline{\mathrm{Eff}}(S)$ is non-polyhedral. □

Corollary B follows now easily:

Corollary 3.5. *Let S be a smooth projective surface endowed with a fibration $f : S \longrightarrow C$. If*

$$\rho(S) > n(f) + 2$$

then $\overline{\mathrm{Eff}}(S)$ is non-polyhedral. In particular, S is not a Mori dream space.

Proof. Theorem 3.4 together with the assumption $\rho(S) > n(f) + 2$ implies that $\overline{\mathrm{Eff}}(S)$ is non-polyhedral. The last claim follows from Theorem 2.2. □

Remark 3.6. If $f : S \longrightarrow C$ is a fibration, then the class e of a general fibre lies on the boundary of $\overline{\mathrm{Eff}}(S)$. Indeed, e is represented by a smooth irreducible curve and $e^2 = 0$, so we can apply [35, Lemma 1.22].

Moreover, if $\rho(S) > n(f) + 2$, then, in the notation of the proof of Proposition 3.3, the affine line

$$e + \mathbb{R}v = H \cap e^{\perp}$$

is a supporting line of the closed disk $\overline{U}$, tangent to its boundary at e. Since

$$\overline{U} \subset \overline{\mathrm{Eff}}(S) \quad \text{and} \quad (e + \mathbb{R}v) \cap \overline{\mathrm{Eff}}(S) = \{e\},$$

the affine section $H \cap \overline{\mathrm{Eff}}(S)$ is not locally polygonal at e. Thus the argument detects the "roundedness", and hence the non-polyhedrality, of $\overline{\mathrm{Eff}}(S)$ locally at e.

Notice, however, that $\mathbb{R}_{\geq 0} \cdot e$ is an extremal ray of $\overline{\mathrm{Eff}}(S)$ if and only if $n(f) = 0$.

Remark 3.7. The converse of Theorem A does not hold: there exist fibrations $f : S \longrightarrow C$ such that $\rho(S) = n(f) + 2$ and nevertheless $\overline{\mathrm{Eff}}(S)$ is non-polyhedral. An explicit example is provided by the Fermat quartic surface $S_4 \subset \mathbb{P}^3$, endowed with the fibration described in Remark 6.6.

We conclude this section with a direct consequence of Theorem 3.4 concerning the Mordell–Weil group of an elliptic fibration.

Corollary 3.8. *Let S be a smooth projective surface with $q(S) = 0$ and let $f : S \longrightarrow C$ be an elliptic fibration with a section. If $\overline{\mathrm{Eff}}(S)$ is polyhedral, then the Mordell–Weil group $\mathrm{MW}(f)$ of the generic fibre of f is finite.*

Proof. By Theorem 3.4, we have $\rho(S) = n(f) + 2$. On the other hand, the Shioda-Tate formula [48, Corollary 6.13] gives

$$\rho(S) = n(f) + 2 + \operatorname{rank} \mathrm{MW}(f),$$

hence $\operatorname{rank} \mathrm{MW}(f) = 0$. Since the Mordell-Weil group is finitely generated [48, Theorem 6.1], it is finite. □

Remark 3.9. After the first version of this article was posted on arXiv, we became aware of several related results and observations:

- C. Casagrande remarked that [10, Corollary 2.16] gives the conclusion of Theorem A under the stronger assumption that S is a Mori dream space;
- when the base of the fibration is $\mathbb{P}^1$, the equality $\rho(S) = n(f)+2$ was already obtained in [12, Remark 2.4];
- as A. Laface pointed out to us, a different proof of Theorem A can be deduced from [12, Proposition 2.3];
- under the additional assumption that $\kappa(S) = 1$, a different proof of Corollary 3.8 was recently given in [39, Proposition 3.1];
- M. Schütt pointed out to us that, even when f does not admit a section, under the remaining hypotheses of Corollary 3.8 the Mordell-Weil group of its relative Jacobian is finite.
- Some connected results on fibred elliptic surfaces has been announced in [31].

4. Irregular examples with non-polyhedral $\overline{\mathrm{Eff}}(S)$: self-products of curves and diagonal double Kodaira surfaces

As a first application of the theory developed above, we get the following result due to Rabindranath [46, Theorem 1.1].

Proposition 4.1. *Let C be a smooth curve with $g(C) \geq 1$. Then $\overline{\mathrm{Eff}}(C \times C)$ is non-polyhedral.*

Proof. If $f : C \times C \longrightarrow C$ is the projection onto one of the factors, then $n(f) = 0$. Moreover, we have $\rho(C \times C) \geq 3$, since the classes of the fibres of the two projections and of the diagonal $\Delta \subset C \times C$ are linearly independent in $N^1(C \times C)$ for $g(C) \geq 1$. So the result follows from Corollary B. □

We can use the same technique to prove that some double Kodaira surfaces (i.e., surfaces endowed with two smooth, non-isotrivial fibrations, see [15]) also have a non-polyhedral pseudo-effective cone. The following example can be found in [43], see also [44] for related constructions. Let C be a smooth curve with $g(C) = 2$ and let $\Delta \subset C \times C$ be the diagonal. Then there exists a G-cover

$$\pi : S \longrightarrow C \times C,$$

where G is an extra-special group of order $2^5 = 32$. The cover π is branched precisely along Δ, with branching order 2. Composing π with the two natural projections of $C \times C$, we get two smooth, non-isotrivial fibrations $f_i : S \longrightarrow C$. The surface S is called a *diagonal double Kodaira surface*.

Proposition 4.2. *Let S be a diagonal double Kodaira surface as described above. Then $\overline{\mathrm{Eff}}(S)$ is non-polyhedral.*

Proof. Writing F_1, F_2 for the fibres of the two fibrations on S, and calling D the ramification locus of π, namely, the smooth curve $D \subset S$ such that $\pi^*\Delta = 2D$, we get

$$F_1^2 = F_2^2 = 0, \quad D^2 = -16, \quad F_1 D = F_2 D = 16, \quad F_1 F_2 = 32.$$

Thus the Gram matrix of the restriction of the intersection form to the subspace of $N^1(S)$ spanned by F_1, F_2, D is the matrix

$$\begin{pmatrix} 0 & 32 & 16 \\ 32 & 0 & 16 \\ 16 & 16 & -16 \end{pmatrix},$$

whose determinant is 32768. So F_1, F_2, D are linearly independent in $N^1(S)$, hence we get $\rho(S) \geq 3$. On the other hand, since $f_1 : S \longrightarrow C$ is a smooth fibration, we have $n(f_1) = 0$. Thus the claim follows from Corollary B. □

Similar examples can be constructed using a smooth curve C of any genus $g \geq 2$ and an extra-special group G of order p^{2g+1}, with p dividing $g+1$; see [16] for more details.

Remark 4.3. Since all the fibrations considered in this section satisfy $n(f) = 0$ and the corresponding surfaces have Picard number at least 3, the argument of Proposition 3.3 shows that the class e of a fibre provides an extremal ray of $\overline{\mathrm{Eff}}(S)$ such that $e^2 = 0$, and that $\overline{\mathrm{Eff}}(S)$ is "round" near this class; see Remark 3.6.

5. Double covers of ruled surfaces and Horikawa surfaces

In this section, we show how to apply Corollary B to double covers of ruled surfaces. Let n be a non-negative integer and consider the Hirzebruch ruled surface $\mathbb{F}_n$, endowed with the ruling $\mathbb{F}_n \longrightarrow \mathbb{P}^1$. We have

$$N^1(\mathbb{F}_n) = \mathbb{R}[F] \oplus \mathbb{R}[C_0],$$

where F is a fibre of the ruling and C_0 is a section with $C_0^2 = -n$, which is unique when $n \geq 1$. We take a double cover

$$\pi : S \longrightarrow \mathbb{F}_n,$$

branched over a smooth curve $B \subset \mathbb{F}_n$ that is not supported on a union of fibres of the ruling. Finally, we consider the fibration $f : S \longrightarrow \mathbb{P}^1$, obtained by composing $\pi : S \longrightarrow \mathbb{F}_n$ with the ruling.

We choose the smooth curve B in order to make the Picard number of the corresponding surface S sufficiently large by applying the following

Lemma 5.1. *Assume that the branch divisor B does not contain C_0 and that B is everywhere tangent to C_0, namely, that the restriction $B|_{C_0}$ yields an effective divisor (possibly the zero divisor) which is 2-divisible in* $\mathrm{Div}(C_0)$. *If $C_0 B = 0$, assume in addition $n > 0$. Then $\rho(S) \geq 3$.*

Proof. The assumption that B is everywhere tangent to C_0 implies that the morphism induced by π on the normalization of π^*C_0 is an étale double cover of C_0. Since $C_0 \simeq \mathbb{P}^1$ has trivial fundamental group, this cover is disconnected. Therefore $\pi^*C_0 = C_1 + C_2$, where C_1, C_2 are smooth rational curves in S and $C_1C_2 = \frac{1}{2}C_0B$. The double cover involution on S exchanges C_1 and C_2, so that

$$C_1^2 = C_2^2 = \frac{1}{2}(C_1 + C_2)^2 - C_1C_2 = C_0^2 - \frac{1}{2}C_0B.$$

If $e = \pi^*F$ is the class of a fibre of $f : S \longrightarrow \mathbb{P}^1$, then the projection formula yields

$$(C_1 + C_2)e = (\pi^*C_0)e = C_0(\pi_*e) = C_0(2F) = 2.$$

So, since the cover involution preserves e, we get $C_1e = C_2e = 1$. Thus the Gram matrix of the restriction of the intersection form to the subspace of $N^1(S)$ spanned by C_1, C_2, e is the matrix

$$\begin{pmatrix} C_0^2 - \frac{1}{2}C_0B & \frac{1}{2}C_0B & 1 \\ \frac{1}{2}C_0B & C_0^2 - \frac{1}{2}C_0B & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

whose determinant is $2(C_0B - C_0^2) = 2(C_0B + n) > 0$. Therefore C_1, C_2, e are linearly independent in $N^1(S)$, and hence $\rho(S) \geq 3$. □

Lemma 5.2. *Let k be an algebraically closed field with* $\mathrm{char}(k) = 0$*, and let $\alpha, \beta \in k$ such that $(\alpha, \beta) \neq (0, 0)$. For all integers $b \geq 2$, the polynomial*

$$p(x) = x^{2b} + \alpha x^{2b-1} + \beta \in k[x]$$

has at least one simple root in k. In particular, $p(x)$ is not a square in $k[x]$.

Proof. If $\beta = 0$ then $p(x) = x^{2b} + \alpha x^{2b-1} = x^{2b-1}(x + \alpha)$ has $-\alpha$ as a simple root. If $\alpha = 0$ then $p(x) = x^{2b} + \beta$ which has only simple roots. Thus we can assume $\alpha \neq 0$ and $\beta \neq 0$. The derivative of $p(x)$ is the polynomial

$$p'(x) = 2bx^{2b-1} + (2b-1)\alpha x^{2b-2} = x^{2b-2}(2bx + (2b-1)\alpha),$$

which has the two distinct roots 0 (with multiplicity $2b-2$) and $\gamma := \frac{(1-2b)\alpha}{2b}$ (with multiplicity 1).

As $p(0) = \beta \neq 0$, the only possible multiple root of $p(x)$ is γ. Moreover, since γ is a simple root of $p'(x)$, it can have multiplicity at most 2 as a root of $p(x)$. Thus $p(x)$ has at least $2b - 2 \geq 2$ simple roots. □

In the sequel, we identify $\mathbb{F}_n$ with a toric variety $\mathbb{C}^4 /\!/ (\mathbb{C}^*)^2$ as follows. Take coordinates t_0, t_1, x_0, x_1 on $\mathbb{C}^4$, and let $(\mathbb{C}^*)^2$ act by the weight matrix

$$\begin{pmatrix} t_0 & t_1 & x_0 & x_1 \\ 1 & 1 & 0 & n \\ 0 & 0 & 1 & 1 \end{pmatrix} \tag{5.1}$$

choosing $(t_0, t_1) \cap (x_0, x_1)$ as the irrelevant ideal. Thus t_0 and t_1 have bidegree $(1, 0)$, whereas x_0 has bidegree $(0, 1)$ and x_1 has bidegree $(n, 1)$.

This identifies the vector spaces $H^0(\mathbb{F}_n, \mathscr{O}_{\mathbb{F}_n}(aF + bC_0))$ with the homogeneous polynomials in the variables t_0, t_1, x_0, x_1 of bidegree (a, b) with respect to the grading defined by the weight matrix (5.1).[1]

For simplicity, we write $H^0(\mathbb{F}_n, \mathscr{O}(a, b))$ instead of $H^0(\mathbb{F}_n, \mathscr{O}_{\mathbb{F}_n}(aF + bC_0))$. So the ruling is given by $H^0(\mathbb{F}_n, \mathscr{O}(1, 0)) = \operatorname{span}(t_0, t_1)$, whereas C_0 is defined by the zero locus of $x_0 \in H^0(\mathbb{F}_n, \mathscr{O}(0, 1))$.

Lemma 5.3. *For every pair of integers $a \geq 0$ and $b \geq 2$, we can choose a branch curve $B \in |\mathscr{O}(2a + 2bn, 2b)|$ so that the following two conditions hold*:

(1) *B is smooth and everywhere tangent to C_0;*
(2) *every fibre of $f : S \longrightarrow \mathbb{P}^1$ is irreducible.*

Proof. Fix integers $a \geq 0$ and $b \geq 2$, and choose pairwise distinct nonzero complex numbers $c_1, \ldots, c_a$. We consider global sections $p_B \in H^0(\mathbb{F}_n, \mathscr{O}(2a + 2bn, 2b))$ of the form

$$p_B = x_1^{2b} \prod_{i=1}^{a} (t_0 - c_i t_1)^2 + x_0 g(t_0, t_1; x_0, x_1), \tag{5.2}$$

where

$$g \in H^0(\mathbb{F}_n, \mathscr{O}(2a + 2bn, 2b - 1)).$$

The zero locus B of p_B is an effective divisor whose class is 2-divisible in $\operatorname{Pic}(\mathbb{F}_n)$, and so there is a double cover

$$\pi : S \longrightarrow \mathbb{F}_n$$

branched along B.[2]

As g varies, the sections in (5.2) form an affine space $\mathscr{L}$, which is an open subset of the projective linear system generated by

$$x_1^{2b} \prod_{i=1}^{a} (t_0 - c_i t_1)^2 \quad \text{and} \quad x_0 H^0(\mathbb{F}_n, \mathscr{O}(2a + 2bn, 2b - 1)).$$

The base points of $\mathscr{L}$ are precisely

$$P_i = [c_i : 1 : 0 : 1], \quad 1 \leq i \leq a.$$

Every $B \in \mathscr{L}$ satisfies

$$B|_{C_0} = 2(P_1 + \cdots + P_a),$$

and is therefore everywhere tangent to C_0. A general member of $\mathscr{L}$ is smooth at the points P_i, and Bertini's theorem shows that it is also smooth away from them. Hence a general member of $\mathscr{L}$ is smooth, and condition (1) is satisfied.[3]

To verify condition (2), it is enough to choose a curve B that intersects every fibre of the ruling transversely at some point. Indeed, an elementary local computation shows that $\pi^* F$ is smooth at the point lying over such an intersection. If $\pi^* F$

[1] For the general theory of homogeneous coordinates on toric varieties, see [17, Chapter 5].

[2] Such a double cover is unique up to isomorphism, since $\operatorname{Pic}(\mathbb{F}_n)$ has no 2-torsion.

[3] If $a = 0$, then B is disjoint from C_0 and the corresponding linear system is base-point free.

were reducible, its two components would meet at that point, contradicting the smoothness of π^*F there.

The locus of pairs $(B,t)\in\mathscr{L}\times\mathbb{P}^1$ such that the restriction of p_B to F_t has no simple zero is closed. Since the projection $\mathscr{L}\times\mathbb{P}^1\longrightarrow\mathscr{L}$ is proper, its image is closed. We now construct a member of $\mathscr{L}$ that does not belong to this image.

Consider the section

$$p_B = x_1^{2b}\prod_{i=1}^{a}(t_0-c_it_1)^2 + t_1^{2a+n}x_0x_1^{2b-1} + t_0^{2a+2bn}x_0^{2b}.$$

We claim that its zero locus intersects every fibre of the ruling transversely in at least one point. Fix $[\bar{t}_0:\bar{t}_1]\in\mathbb{P}^1$. The restriction to the corresponding fibre is

$$p_B(\bar{t}_0,\bar{t}_1;x_0,x_1) = \tau x_1^{2b} + \alpha x_0x_1^{2b-1} + \beta x_0^{2b}, \tag{5.3}$$

where

$$\tau = \prod_{i=1}^{a}(\bar{t}_0-c_i\bar{t}_1)^2, \quad \alpha = \bar{t}_1^{2a+n}, \quad \beta = \bar{t}_0^{2a+2bn}.$$

Proving our claim is equivalent to showing that the binary form (5.3) has a linear factor of multiplicity 1.

Since all the c_i are nonzero, at most one of α,β,τ vanishes.

If $\beta=0$, then $\alpha\tau\neq 0$ and

$$p_B = x_1^{2b-1}(\tau x_1+\alpha x_0),$$

so the factor $\tau x_1+\alpha x_0$ has multiplicity 1.

Suppose now that $\beta\neq 0$. Setting $x=x_1/x_0$ gives

$$p(x) = \tau x^{2b} + \alpha x^{2b-1} + \beta.$$

If $\tau\neq 0$, after dividing by τ we may apply Lemma 5.2, which shows that $p(x)$ has a simple root. If $\tau=0$, then $\alpha\beta\neq 0$ and

$$p(x) = \alpha x^{2b-1} + \beta,$$

which has only simple roots. This proves our claim: every fibre meets B transversely in at least one point.

It follows that condition (2) holds on a nonempty open subset of $\mathscr{L}$. Since $\mathscr{L}$ is irreducible, this open subset intersects the nonempty open subset parametrizing the smooth members. Choosing B in their intersection proves the lemma. □

We can now state the main result of this section.

Theorem 5.4. *Let $a,n\geq 0$ and $b\geq 2$ be integers, and assume that $a+n\geq 2$. Consider the double cover $\pi\colon S\longrightarrow\mathbb{F}_n$, branched along a curve $B\in|2(a+bn)F+2bC_0|$ whose defining polynomial is general among those in the affine space $\mathscr{L}$ defined above. Then S is a smooth minimal surface such that $\overline{\mathrm{Eff}}(S)$ is non-polyhedral.*

Proof. By Lemma 5.3, a general branch curve $B \in \mathscr{L}$ is smooth, and hence the corresponding double cover S is smooth. Moreover, every fibre of $f\colon S \longrightarrow \mathbb{P}^1$ is irreducible, so $n(f) = 0$. We have

$$C_0 B = C_0\,(2(a+bn)F + 2bC_0) = 2a.$$

Thus, if $C_0 B = 0$ then $a = 0$ and the assumption $a + n \geq 2$ implies $n \geq 2$. Therefore Lemma 5.1 applies and gives $\rho(S) \geq 3$. Consequently, Corollary B implies that $\overline{\mathrm{Eff}}(S)$ is non-polyhedral.

Finally, the double cover formula gives

$$K_S = \pi^*\left(K_{\mathbb{F}_n} + (a+bn)F + bC_0\right),$$

where the divisor inside the pullback can be written as

$$(b-2)(C_0 + nF) + (a+n-2)F.$$

Since F and $C_0 + nF$ are nef and both coefficients are non-negative, this divisor is nef. This means that K_S is also nef, hence S is minimal. □

Let us recall that a *Horikawa surface* is a minimal surface S of general type satisfying $K_S^2 = 2p_g(S) - 4$.[4] Horikawa surfaces satisfy $p_g(S) \geq 3$ and $q(S) = 0$; we refer to [3, Chapter VII] and to the references therein for further details.

Corollary 5.5. *For every integer $p \geq 3$, there exists a Horikawa surface S with $p_g(S) = p$ such that $\overline{\mathrm{Eff}}(S)$ is non-polyhedral. In particular, S is not a Mori dream space.*

Proof. Set $b = 3$ and choose a, n and a general branch curve B as in Theorem 5.4. Writing the linear equivalence class of B as

$$B = 2L, \quad L = (a+3n)F + 3C_0,$$

the double cover formulas give

$$p_g(S) = h^0(\mathbb{F}_n, K_{\mathbb{F}_n} + L) = 2a + 3n - 2$$

and

$$K_S^2 = 2(K_{\mathbb{F}_n} + L)^2 = 4a + 6n - 8 = 2p_g(S) - 4.$$

Moreover, K_S is nef by Theorem 5.4, and it is big whenever $2a + 3n \geq 5$.

Now fix an integer $p \geq 3$. If p is odd, take

$$n = 1, \quad a = (p-1)/2$$

whereas if p is even, take

$$n = 0, \quad a = (p+2)/2.$$

In both cases, $a + n \geq 2$, $2a + 3n = p + 2 \geq 5$, and $p_g(S) = p$. Hence S is a Horikawa surface. Its pseudo-effective cone is non-polyhedral by Theorem 5.4, so S is not a Mori dream space; see also [28, Theorem 1.6]. □

[4] That is, Horikawa surfaces attain equality in the Noether inequality $K_S^2 \geq 2p_g(S) - 4$.

6. Surfaces in $\mathbb{P}^3$ containing lines and Fermat surfaces

A natural class of surfaces to which one may try to apply Corollary B is the class of smooth surfaces $S \subset \mathbb{P}^3$ containing a line ℓ.

Let H be the hyperplane class of such a surface S. The residual pencil $|H-\ell|$ has no fixed components, because the only curve common to two distinct planes containing ℓ is ℓ itself.

Set $d := \deg S \geq 2$. By adjunction, we have $K_S = (d-4)H$. Thus, $K_S\ell = d-4$, and the genus formula gives $\ell^2 = 2-d$. Therefore,

$$(H-\ell)^2 = H^2 - 2H\ell + \ell^2 = d - 2 + (2-d) = 0.$$

Since the pencil has no fixed components, this equality implies that it has no base points. Hence, it defines a morphism $f\colon S \longrightarrow \mathbb{P}^1$. Its fibres are plane curves of degree $d-1$ and are therefore connected, so f is a fibration.

Remark 6.1. The fibres of f are always reduced. Indeed, let $\Pi \subset \mathbb{P}^3$ be a plane containing ℓ, and suppose that the corresponding fibre contains an irreducible component Γ with multiplicity greater than one. At a general smooth point $p \in \Gamma$, the plane Π coincides with the tangent plane to S at p, namely, Π is tangent to S along Γ. It follows that the Gauss map $\gamma\colon S \longrightarrow (\mathbb{P}^3)^*$ contracts Γ to a point, contradicting the fact that γ is finite [53].

In order to estimate $n(f)$, it is useful to recall some standard facts about Milnor numbers of plane curves.

Definition 6.2. Let $C \subset \mathbb{P}^2$ be a reduced curve of degree $d-1$. Writing $\chi_{\mathrm{top}}(C)$ for its Euler characteristic, we define the *total Milnor number of* C as

$$\mu(C) := \chi_{\mathrm{top}}(C) + (d-1)(d-4).$$

By the standard theory of plane curves, $\mu(C) \geq 0$ and C is smooth if and only if $\mu(C) = 0$. The number $\mu(C)$ measures the difference between the Euler characteristic of C and that of a smooth plane curve of the same degree.

Write $C = \sum_{i=1}^k C_i$, where the C_i are the irreducible components of C. Let $\nu\colon \tilde{C} \to C$ be the normalization of C and, for all $p \in C$, write $r_p := \#\nu^{-1}(p)$ for the number of local branches at p. Then

$$\chi_{\mathrm{top}}(C) = \sum_{i=1}^{k}(2-2g_i) - \sum_{p\in \mathrm{Sing}(C)}(r_p - 1)$$

where g_i is the genus of the normalization of C_i.

The natural inclusion $\mathscr{O}_C \hookrightarrow \nu_*\mathscr{O}_{\tilde{C}}$ has cokernel sheaf $\mathscr{Q}$ supported on $\mathrm{Sing}(C)$, so we can write $\mathscr{Q} = \oplus_{p\in\mathrm{Sing}(C)}\mathscr{Q}_p$ with $\mathscr{Q}_p$ supported on p. Setting $\delta_p := \text{length } \mathscr{Q}_p$, the residue map of the normalization implies

$$\delta_p \geq r_p - 1.$$

Moreover, from the short exact sequence

$$0 \to \mathcal{O}_C \to \nu_* \mathcal{O}_{\tilde{C}} \to \mathcal{Q} \to 0$$

we obtain the formula

$$p_a(C) = p_a(\tilde{C}) + \sum_{p \in \mathrm{Sing}(C)} \delta_p = \sum_{i=1}^{k} g_i - (k-1) + \sum_{p \in \mathrm{Sing}(C)} \delta_p. \tag{6.1}$$

Then

$$\mu(C) = \chi_{\mathrm{top}}(C) - (2 - 2p_a(C)) = \sum_{p \in \mathrm{Sing}(C)} \mu_p(C) \geq \sum_{p \in \mathrm{Sing}(C)} \delta_p, \tag{6.2}$$

where $\mu_p(C) := 2\delta_p - r_p + 1$ equals the *local Milnor number* of C at p, cf. [37, Theorem 10.5].

Lemma 6.3. *Let C be a reduced plane curve of degree $d-1$ with k irreducible components. Then*

$$\mu(C) \geq \frac{1}{2}(d-1)(k-1).$$

Proof. Setting $d_i := \deg C_i$, so that $d - 1 = \sum_{i=1}^{k} d_i$, by (6.1) we get

$$\begin{aligned} \sum_{p \in \mathrm{Sing}(C)} \delta_p &= \frac{1}{2}(d-2)(d-3) - \sum_{i=1}^{k} g_i + k - 1 \\ &\geq \frac{1}{2}(d-2)(d-3) - \frac{1}{2}\sum_{i=1}^{k}(d_i - 1)(d_i - 2) + k - 1 = \frac{1}{2}\sum_{i \neq j} d_i d_j. \end{aligned} \tag{6.3}$$

Since

$$\frac{1}{2}\sum_{i \neq j} d_i d_j = \frac{1}{2}\sum_{i=1}^{k} d_i(d - 1 - d_i) \geq \frac{1}{2}\sum_{i=1}^{k} d_i(k-1) = \frac{1}{2}(d-1)(k-1), \tag{6.4}$$

the claim now follows from (6.2), (6.3) and (6.4). □

We are now in a position to prove the following

Theorem 6.4. *Let $S \subset \mathbb{P}^3$ be a smooth surface of degree $d \geq 2$ containing a line ℓ. If*

$$\rho(S) > \frac{2(d-2)^2(d+2)}{d-1} + 2$$

then $\overline{\mathrm{Eff}}(S)$ is non-polyhedral, and so S is not a Mori dream space.

Proof. We consider the fibration $f : S \to \mathbb{P}^1$ given by $|H - \ell|$. For all $t \in \mathbb{P}^1$, denote by F_t the corresponding fibre. All F_t are plane curves of degree $d-1$. We also denote by F a general fibre.

Since F is a smooth plane curve of degree $d-1$, the definition of $\mu(F_t)$ gives

$$\mu(F_t) = \chi_{\mathrm{top}}(F_t) - \chi_{\mathrm{top}}(F),$$

and so, by the Zeuthen-Segre formula [3, Chapter III, Proposition 11.4 and Remark 11.5], we get

$$\sum_{t\in\mathbb{P}^1}\mu(F_t) = \chi_{\text{top}}(S) - \chi_{\text{top}}(\mathbb{P}^1)\chi_{\text{top}}(F)$$
$$= d^3 - 4d^2 + 6d - 2(2-(d-2)(d-3)) = (d-2)^2(d+2).$$

Furthermore, all fibres F_t are reduced, see Remark 6.1. Therefore we can apply Lemma 6.3, obtaining

$$\sum_{t\in\mathbb{P}^1}\mu(F_t) \geq \frac{1}{2}(d-1)n(f)$$

and so

$$n(f) \leq \frac{2(d-2)^2(d+2)}{d-1}. \tag{6.5}$$

The claim now follows from Corollary B. □

We apply Theorem 6.4 to the *Fermat surface* $S_d \subset \mathbb{P}^3$, defined by

$$x_0^d - x_1^d + x_2^d - x_3^d = 0.$$

We can take the line $\ell \subset S_d$ of equation

$$x_0 - x_1 = x_2 - x_3 = 0$$

and consider the corresponding fibration

$$f_d : S_d \longrightarrow \mathbb{P}^1,$$

whose fibres F_t are the members of the movable part of the linear system cut out on S_d by the pencil of planes

$$x_0 - x_1 = t(x_2 - x_3), \quad t \in \mathbb{P}^1$$

passing through ℓ.

Theorem 6.5. *Let $S_d \subseteq \mathbb{P}^3$ be a Fermat surface of degree $d \geq 4$. Then $\overline{\text{Eff}}(S_d)$ is non-polyhedral, and so S_d is not a Mori dream space.*

Proof. By [51, formula 5], since all the additional terms appearing there are non-negative, we have

$$\rho(S_d) \geq 3(d-1)(d-2)+1.$$

Thus, by Theorem 6.4, if

$$3(d-1)(d-2)+1 > 2\frac{(d-2)^2(d+2)}{d-1} + 2 \tag{6.6}$$

then the pseudo-effective cone of S_d is non-polyhedral. After multiplying by $d-1$, (6.6) is equivalent to

$$(d-3)(d^2-5d+7) > 0,$$

which is true for all $d \geq 4$. □

The degree bound in Theorem 6.5 is sharp: the Fermat surfaces S_2 and S_3 are del Pezzo surfaces and hence Mori dream spaces.

Remark 6.6. It was already known that the pseudo-effective cone $\overline{\mathrm{Eff}}(S_4)$ of the Fermat quartic $S_4 \subset \mathbb{P}^3$ is non-polyhedral. In fact, S_4 is a K3 surface with Picard number 20, and a result of Shioda and Inose [52, Theorem 5 in §5] (see also [49, §§17–18]) shows that its automorphism group is infinite. Moreover, S_4 contains lines, which are (-2)-curves. Therefore, by results of Pjateckiĭ-Šapiro and Šafarevič, and of Sterk [30, Chapter 8, §4, Corollary 4.7], the surface S_4 contains infinitely many (-2)-curves, which provide infinitely many extremal rays of its pseudo-effective cone.

There is another interesting fibration $g\colon S_4 \longrightarrow \mathbb{P}^1$ defined by the rational function

$$t = \frac{x_0^2 - x_1^2}{x_2^2 - x_3^2} = -\frac{x_2^2 + x_3^2}{x_0^2 + x_1^2}.$$

This is an elliptic fibration having six singular fibres, each a cycle of four lines of Kodaira type I_4 [45, Example p. 44]. Thus, we have

$$n(g) + 2 = 6 \cdot (4-1) + 2 = 20 = \rho(S_4).$$

This shows at once two interesting things:

(1) the equality $\rho(S) = n(f) + 2$ in Theorem A is necessary, but not sufficient, for the polyhedrality of $\overline{\mathrm{Eff}}(S)$;
(2) if $\overline{\mathrm{Eff}}(S)$ is non-polyhedral, then S may admit two distinct fibrations with different values of the invariant $n(f)$. In fact, for the Fermat surface $S_4 \subset \mathbb{P}^3$ we showed that $n(g) = 18$; on the other hand, we have $n(f_4) \leq 16$, see (6.5). By Theorem A, this situation cannot occur if $\overline{\mathrm{Eff}}(S)$ is polyhedral.

7. Product-quotient surfaces

If C is a smooth curve with $g(C) \geq 1$, we have seen (Proposition 4.1) that $\overline{\mathrm{Eff}}(C \times C)$ is non-polyhedral. However, since $q(C \times C) = 2g(C) > 0$, we do not need to study its pseudo-effective cone in order to conclude that $C \times C$ is not a Mori dream space.

To obtain regular examples, we slightly modify the setting and consider minimal resolutions of quotients of products of curves by diagonal actions of finite groups. Such surfaces have been extensively studied in the literature [4–7, 13, 23–25, 38, 40, 41] and are known as *product-quotient surfaces*.

Definition 7.1. Let C_1, C_2 be two smooth projective curves and let G be a finite group acting faithfully on both of them. A *product-quotient* surface is the minimal resolution of singularities $\lambda\colon S \longrightarrow X$ of the *quotient model* $X = (C_1 \times C_2)/G$, where G acts diagonally on the product.

The surface S comes naturally with two fibrations

$$f_1\colon S \longrightarrow C_1/G, \quad f_2\colon S \longrightarrow C_2/G,$$

whose general fibres we denote by F_1 and F_2, respectively.

We refer to [50, Theorem 2.1], where the author studied the fibres of f_1 and f_2. In particular, one can easily observe that

$$F_1 \simeq C_2, \quad F_2 \simeq C_1 \quad \text{and} \quad F_1F_2 = |G|,$$

and both fibrations f_1 and f_2 are *isotrivial*. Moreover

$$n(f_1) = n(f_2) = l(X) \tag{7.1}$$

where $l(X)$ is the number of irreducible curves contracted by $\lambda\colon S \longrightarrow X$, see [8, Definition 2.3]. Indeed, the reduced support of every reducible fibre consists of one non-exceptional component (called the *central component*) together with a finite number of Hirzebruch-Jung strings contracted by λ to singular points of X. Since every exceptional curve occurs in exactly one fibre, summing the numbers of components other than the central components gives (7.1).

Moreover, by [50, Proposition 2.2] we get

Proposition 7.2. *The irregularity of a product-quotient surface S is*

$$q(S) = g(C_1/G) + g(C_2/G).$$

7.1. **Surfaces isogenous to a product of unmixed type.** We first consider the case where $g(C_1), g(C_2) \geq 2$ and the action of G on $C_1 \times C_2$ is free. In this situation, X is smooth and

$$S = X = (C_1 \times C_2)/G$$

is called a *surface isogenous to a product of unmixed type*.

Because of the assumption on $g(C_i)$, the canonical divisor $K_{C_1\times C_2}$ is ample. Since the quotient map $C_1 \times C_2 \longrightarrow S$ is finite and étale, the canonical divisor K_S is ample as well. Consequently, S is a minimal surface of general type.

Proposition 7.3. *Let $S = (C_1 \times C_2)/G$ be a surface isogenous to a product of unmixed type. If $\rho(S) \geq 3$, then $\overline{\mathrm{Eff}}(S)$ is non-polyhedral. In particular, S is not a Mori dream space.*

Proof. Since the action of G on $C_1 \times C_2$ is free, by [50, Theorem 2.1] all the singular fibres of f_i are multiples of smooth curves, hence $n(f_i) = 0$ and the claim follows from Corollary B. □

Recall that a *non-trivial correspondence* between C_1 and C_2 is a possibly reducible curve $\Gamma \subset C_1 \times C_2$ such that both natural projections

$$\Gamma \longrightarrow C_1, \quad \Gamma \longrightarrow C_2$$

are surjective.

Proposition 7.4. *Let $S = (C_1 \times C_2)/G$ be a surface isogenous to a product of unmixed type. Assume that there exists a non-trivial G-invariant correspondence $\Gamma \subset C_1 \times C_2$ such that $\Gamma^2 \leq 0$. Then $\overline{\mathrm{Eff}}(S)$ is non-polyhedral.*

Proof. Let

$$\pi\colon C_1 \times C_2 \longrightarrow S = (C_1 \times C_2)/G$$

be the quotient map and set $B = \pi(\Gamma)$. Since Γ is G-invariant and π is unramified, we get $\pi^* B = \Gamma$. We claim that the three classes

$$F_1,\ F_2,\ B$$

are linearly independent in $N^1(S)$. In fact, suppose that there exists a linear relation

$$\lambda_1 F_1 + \lambda_2 F_2 = \mu B,$$

where $\lambda_1, \lambda_2, \mu \in \mathbb{R}$ and we can assume $\mu \geq 0$. Pulling back to $C_1 \times C_2$ via π, we get

$$\tau_1 C_2 + \tau_2 C_1 = \mu\Gamma,$$

where $\tau_1 = \lambda_1 \cdot |G|$, $\tau_2 = \lambda_2 \cdot |G|$. Intersecting with C_1 we get $\tau_1 \geq 0$, and intersecting with C_2 we get $\tau_2 \geq 0$. Thus, intersecting with Γ we get

$$0 \leq (\tau_1 C_2 + \tau_2 C_1)\Gamma = \mu\Gamma^2 \leq 0.$$

Then $(\tau_1 C_2 + \tau_2 C_1)\Gamma = 0$. As Γ is a non-trivial correspondence, both products $C_1\Gamma$ and $C_2\Gamma$ are strictly positive. Then $\tau_1 = \tau_2 = \mu = 0$ and so $\lambda_1 = \lambda_2 = 0$.

Thus F_1, F_2, and B are linearly independent, hence $\rho(S) \geq 3$. The conclusion now follows from Proposition 7.3. □

Remark 7.5. In Proposition 7.4, the condition that Γ be G-invariant is essential. Indeed, if $C_1 = C_2 = C$, then the diagonal $\Delta \subset C \times C$ is a non-trivial correspondence having self-intersection $\Delta^2 = 2 - 2g(C)$, which is negative whenever $g(C) \geq 2$.

However, the actions of G on the two copies of C need not coincide, and in this case Δ need not be G-invariant. There are examples of free diagonal G-actions for which this happens and the resulting surface $S = (C \times C)/G$ is a Mori dream space. One such example is the Beauville surface described in Remark 7.8.

We now give some examples of surfaces to which Proposition 7.4 may be applied. The first example is due to J. Starr, see [42].

Example (J. Starr) 7.6. Let $p \geq 7$ be a prime number and consider the finite group $G = \mathbb{Z}_p \times \mathbb{Z}_p$. The three elements

$$g_1 = (1, 0), \quad g_2 = (0, -1), \quad g_3 = (-1, 1)$$

generate G and satisfy $g_1 + g_2 + g_3 = 0$, hence there is an associated G-cover

$$\pi\colon C \longrightarrow \mathbb{P}^1,$$

branched over $0, 1, \infty$ such that

- the stabilizer of every point lying over 0 is $\langle g_1 \rangle$;
- the stabilizer of every point lying over 1 is $\langle g_2 \rangle$;
- the stabilizer of every point lying over ∞ is $\langle g_3 \rangle$.

If $x \in C$ and $g \in G$, we denote the action of g on x by $g \cdot x$, keeping the additive notation so that

$$g \cdot (h \cdot x) = (g + h) \cdot x.$$

By the Riemann-Hurwitz formula, $g(C) = \frac{1}{2}(p-1)(p-2)$. Now, let us write $\mathbb{F}_p$ for the field with p elements, whose additive group is $\mathbb{Z}_p$, and consider the automorphism $\phi\colon G \longrightarrow G$ given by the matrix

$$\begin{pmatrix} 2 & -3 \\ 1 & -2 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{F}_p).$$

We have

$$\phi(g_1) = (2,1), \quad \phi(g_2) = (3,2), \quad \phi(g_3) = (-5,-3)$$

and so, since $p \geq 7$,

$$\langle g_i \rangle \cap \langle \phi(g_j) \rangle = \{(0,0)\} \quad \text{for all } i, j.$$

Therefore the action of G on $C \times C$ defined by

$$g \cdot (x, y) := (g \cdot x, \phi(g) \cdot y)$$

is free, and we obtain a surface isogenous to a product of unmixed type

$$S = (C \times C)/G,$$

which is a minimal surface of general type and satisfies $q(S) = 0$ by Proposition 7.2.

Moreover, as will be relevant in the sequel, 1 is an eigenvalue of ϕ, and we have

$$\ker(\phi - \mathrm{id}) = \langle (3,1) \rangle, \quad \mathrm{im}(\phi - \mathrm{id}) = \langle (1,1) \rangle.$$

For any $g \in G$, we can consider the graph of the automorphism $x \mapsto g \cdot x$, i.e.

$$\Gamma_g = \{(x, g \cdot x) \mid x \in C\} \subset C \times C.$$

It is a smooth divisor isomorphic to C and such that $\Gamma_g^2 = 2 - 2g(C)$. Note that $\Gamma_{(0,0)}$ is the diagonal $\Delta \subset C \times C$, and that

$$\Gamma_{h_1} \cap \Gamma_{h_2} \neq \emptyset \quad \text{if and only if} \quad h_1 - h_2 \in \langle g_1 \rangle \cup \langle g_2 \rangle \cup \langle g_3 \rangle. \tag{7.2}$$

Moreover, for all $g \in G$, we have

$$\begin{aligned} g \cdot \Delta &= \{(g \cdot x, \phi(g) \cdot x) \mid x \in C\} \\ &= \{(y, (\phi(g) - g) \cdot y) \mid y \in C\} = \Gamma_{\phi(g) - g}. \end{aligned}$$

Theorem 7.7. *For all prime numbers $p \geq 7$, the pseudo-effective cone $\overline{\mathrm{Eff}}(S)$ of the surface S constructed above is non-polyhedral. In particular, S is not a Mori dream space.*

Proof. Let Γ be the reduced G-orbit of the diagonal. Since $\ker(\phi - \mathrm{id})$ is precisely the stabilizer of Δ, we have

$$\Gamma = \sum_{\bar{g} \in G/\ker(\phi - \mathrm{id})} g \cdot \Delta = \sum_{h \in \mathrm{im}(\phi - \mathrm{id})} \Gamma_h.$$

In particular, Γ is a G-invariant non-trivial correspondence.

If $h_1, h_2 \in \mathrm{im}(\phi - \mathrm{id})$, then also $h_1 - h_2 \in \mathrm{im}(\phi - \mathrm{id})$. Since $\mathrm{im}(\phi - \mathrm{id})$ intersects each stabilizer $\langle g_i \rangle$ only at the identity, by (7.2) we get

$$\Gamma_{h_1}\Gamma_{h_2} = 0 \quad \text{whenever} \quad h_1 \neq h_2.$$

Since $\mathrm{im}(\phi - \mathrm{id})$ has order p, we obtain

$$\Gamma^2 = \left(\sum_{h \in \mathrm{im}(\phi - \mathrm{id})} \Gamma_h \right)^2 = \sum_{h \in \mathrm{im}(\phi - \mathrm{id})} \Gamma_h^2 = p(2 - 2g(C)) < 0,$$

and the claim follows from Proposition 7.4. □

Remark 7.8. For primes $p \leq 5$, it is not possible to find a matrix $A \in \mathrm{GL}_2(\mathbb{F}_p)$ satisfying all the conditions imposed in the example above. Indeed, set

$$\Sigma := \{\langle g_1 \rangle, \langle g_2 \rangle, \langle g_3 \rangle\} \subset \mathbb{P}^1(\mathbb{F}_p).$$

The diagonal action of G on $C \times C$ is free precisely when

$$A(\Sigma) \cap \Sigma = \varnothing.$$

For $p \leq 3$, this is impossible, as we can see by counting the elements of $\mathbb{P}^1(\mathbb{F}_p)$.

For $p = 5$, a free diagonal action does exist and yields the classical surface with

$$p_g = q = 0, \quad K^2 = 8,$$

described by Beauville in [9, Ex. X.13(4)]. In this case, however,

$$A(\Sigma) = \mathbb{P}^1(\mathbb{F}_5) \setminus \Sigma,$$

so A fixes no line in $\mathbb{F}_5^2$.[5] In particular, 1 is not an eigenvalue of A, and the construction used in the preceding example does not apply. Indeed, Beauville's surface is a Mori dream space, see [33, Proposition 3.4].

More examples of surfaces to which Proposition 7.4 may be applied can be constructed as follows.

Proposition 7.9. *Let $S = (C_1 \times C_2)/G$ be a surface isogenous to a product of unmixed type, let H be a normal subgroup of G and set $Q := G/H$.*

Assume that

$$\eta_i \colon C_i/H \longrightarrow C_i/G, \quad i = 1, 2$$

are isomorphic as Q-covers, and that

$$g(C_1/H) = g(C_2/H) \geq 1.$$

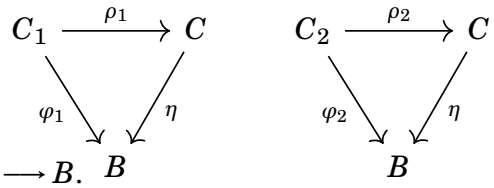
Then $\overline{\mathrm{Eff}}(S)$ is non-polyhedral, and so S is not a Mori dream space.

[5] Here the six points of $\mathbb{P}^1(\mathbb{F}_5)$ parametrize the six subgroups of order 5 of $G = \mathbb{Z}_5 \times \mathbb{Z}_5$. The three stabilizer subgroups $\langle g_i \rangle$ form one half, and their images under A form the other half. Since A exchanges the two halves, it cannot admit an eigenline.

Proof. After choosing an isomorphism of the two Q-covers, we identify

$$B := C_1/G = C_2/G, \quad C := C_1/H = C_2/H, \quad \eta := \eta_1 = \eta_2.$$

By assumption there are two factorizations



of the G-covers $\varphi_i : C_i \longrightarrow B$.

The $(H \times H)$-cover

$$\rho = \rho_1 \times \rho_2 : C_1 \times C_2 \longrightarrow C \times C$$

is equivariant with respect to the quotient homomorphism $G \longrightarrow Q$: explicitly, if $\bar{g}$ denotes the image of $g \in G$ in Q, then

$$\rho(g \cdot (x, y)) = \bar{g} \cdot \rho(x, y).$$

Thus there is a commutative diagram

$$\begin{array}{ccc} C_1 \times C_2 & \xrightarrow{\rho} & C \times C \\ \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \bar{\pi}} \\ S = (C_1 \times C_2)/G & \xrightarrow[\bar{\rho}]{} & (C \times C)/Q, \end{array}$$

where Q acts diagonally on $C \times C$.

Since the action of Q on the two factors of $C \times C$ is the same, the diagonal $\Delta \subset C \times C$ is Q-invariant. We set

$$\Gamma := \rho^*(\Delta) \subset C_1 \times C_2, \quad R = \bar{\pi}(\Delta) \subset (C \times C)/Q,$$

and we note that Γ is a non-trivial correspondence in $C_1 \times C_2$.

Since both $\bar{\pi}$ and $\bar{\rho}$ are finite morphisms between normal varieties, the pull-back of Weil divisors is well-defined; see for instance [19, Lemma 1.5].

Moreover, the action of Q on C is faithful, so the same is true for its restriction to $\Delta \simeq C$. Therefore $\bar{\pi}$ is unramified at the generic point of Δ. Since Δ is the unique prime divisor lying over R, it follows that

$$\bar{\pi}^* R = \Delta.$$

Summing up, we get

$$\Gamma = \rho^*(\Delta) = \rho^*(\bar{\pi}^* R) = \pi^*(\bar{\rho}^* R),$$

that is, the correspondence Γ is G-invariant. Finally, since $g(C) \geq 1$ we have $\Delta^2 \leq 0$ and consequently $\Gamma^2 \leq 0$, so we can conclude using Proposition 7.4. □

We use Proposition 7.9 to construct other examples of minimal surfaces of general type with $q = 0$ that are not Mori dream spaces. The case $p = 3$ was explained to us by W. Sawin in [42].

Example (W. Sawin) 7.10. Let $p \geq 3$ be a prime number and take three homogeneous polynomials $f(x_0, x_1), g(x_0, x_1), h(x_0, x_1)$ of degree p such that the product fgh has $3p$ distinct roots in $\mathbb{P}^1$.

Take $\mathbb{P}^3$ with homogeneous coordinates x_0, x_1, y_0, y_1 and let $C_1 \subset \mathbb{P}^3$ be the curve

$$\begin{cases} y_0^p = f \\ y_1^p = g \end{cases}$$

The curve C_1 is smooth by the assumption on fgh. We consider the faithful action of $G := \mathbb{Z}_p \times \mathbb{Z}_p$ on C_1 induced by the following action on $\mathbb{P}^3$:

$$(a,b)\cdot(x_0, x_1, y_0, y_1) = (x_0, x_1, \xi^a y_0, \xi^b y_1) \tag{7.3}$$

where $\xi = e^{2\pi i/p}$. The induced G-cover $\varphi_1 \colon C_1 \longrightarrow C_1/G \simeq \mathbb{P}^1$ is given by $(x_0, x_1, y_0, y_1) \mapsto (x_0, x_1)$.

Now take the weighted projective space $\mathbb{P}(1, 1, 1, 2)$ with weighted coordinates x_0, x_1, z_0, z_1; here z_1 is the variable of weight 2. Let $C_2 \subset \mathbb{P}(1, 1, 1, 2)$ be the curve

$$\begin{cases} z_0^p = h \\ z_1^p = fg \end{cases}$$

The curve C_2 is smooth by the assumption on fgh. We consider the action of G on C_2 induced by the following action on $\mathbb{P}(1, 1, 1, 2)$:

$$(a, b)\cdot(x_0, x_1, z_0, z_1) = (x_0, x_1, \xi^{a-b} z_0, \xi^{a+b} z_1) \tag{7.4}$$

Since $p \geq 3$, the action (7.4) is faithful on C_2, so we obtain a further G-cover $\varphi_2 \colon C_2 \longrightarrow C_2/G \simeq \mathbb{P}^1$ given by $(x_0, x_1, z_0, z_1) \mapsto (x_0, x_1)$.[6]

Let H be the cyclic subgroup of G generated by $(1, -1)$.

The H-invariant subring of $\mathbb{C}[x_0, x_1, y_0, y_1]/(y_0^p - f, y_1^p - g)$, with respect to the action induced by (7.3), is generated by x_0, x_1, and $y_0 y_1$. Setting $y := y_0 y_1$ we can write C_1/H as the smooth curve $\{y^p = fg\} \subset \mathbb{P}(1, 1, 2)$ and the cover $\eta_1 \colon C_1/H \longrightarrow \mathbb{P}^1$ is given by $(x_0, x_1, y) \mapsto (x_0, x_1)$.

The H-invariant subring of $\mathbb{C}[x_0, x_1, z_0, z_1]/(z_0^p - h, z_1^p - fg)$, with respect to the action induced by (7.4), is generated by x_0, x_1 and z_1. We can write C_2/H as the smooth curve $\{z_1^p = fg\} \subset \mathbb{P}(1,1,2)$ and the cover $\eta_2 \colon C_2/H \longrightarrow \mathbb{P}^1$ is given by $(x_0, x_1, z_1) \mapsto (x_0, x_1)$.

The homomorphism

$$G \longrightarrow \mathbb{Z}_p, \qquad (a,b) \longmapsto a + b$$

has kernel H, and hence identifies $Q = G/H$ with $\mathbb{Z}_p$. Under this identification, the induced actions on C_1/H and C_2/H are given respectively by

$$y \longmapsto \xi^{a+b} y, \qquad z_1 \longmapsto \xi^{a+b} z_1.$$

Consequently, the identification $y = z_1$ is Q-equivariant, and therefore η_1 and η_2 are isomorphic as Q-covers.

[6] If $p = 2$ then the element $(1, 1) \in G$ would act trivially on C_2.

Theorem 7.11. *For all prime numbers $p \geq 3$, consider the actions of G on C_1 and C_2 induced by (7.3) and (7.4), respectively. Let $S := (C_1 \times C_2)/G$, where G acts diagonally. Then S is a minimal surface of general type isogenous to a product of unmixed type, with $q(S) = 0$, such that $\overline{\mathrm{Eff}}(S)$ is non-polyhedral. In particular, S is not a Mori dream space.*

Proof. Since x_0 and x_1 do not vanish simultaneously on C_1, a standard computation using (7.3) shows that the elements of G fixing some point of C_1 are exactly those of the form $(a, 0)$ or $(0, b)$.

By (7.4), since x_0 and x_1 do not vanish simultaneously on C_2, the only element of this form fixing a point of C_2 is $(0, 0)$. Thus the diagonal action of G on $C_1 \times C_2$ is free, and S is isogenous to a product of unmixed type. Moreover, $q(S) = 0$ by Proposition 7.2.

Furthermore, we showed above that $\eta_i : C_i/H \longrightarrow \mathbb{P}^1$ are isomorphic as G/H-covers. They have degree p and are totally ramified over the $2p$ distinct zeros of fg. Thus the Riemann-Hurwitz formula gives

$$2g(C_i/H) - 2 = -2p + 2p(p-1),$$

hence $g(C_i/H) = (p-1)^2$. This also shows that $g(C_i) \geq 2$, hence $S = (C_1 \times C_2)/G$ is a minimal surface of general type.

Finally, the claim about $\overline{\mathrm{Eff}}(S)$ follows from Proposition 7.9. □

7.2. **Product-quotient surfaces with geometric genus zero.** In [8, Definitions 2.1-2.3] the authors associate to every surface X whose singular points are cyclic quotient singularities a rational number $\gamma(X)$ that depends only on the singularities of X. Then they prove the following result, see [8, Prop. 4.2]:

Proposition 7.12. *If $X = (C_1 \times C_2)/G$ is the quotient model of a product-quotient surface S, then $\gamma(X) \in \mathbb{Z}$ and $p_g(S) + \gamma(X) \geq 0$. Furthermore, denoting by $f_i : S \longrightarrow C_i/G$ the two natural fibrations, we have*

$$h^{1,1}(S) - n(f_i) - 2 = 2(p_g(S) + \gamma(X)).$$

Thus, in this case, Theorem A has the following reformulation.

Corollary 7.13. *Let S be a product-quotient surface with quotient model X. If $\overline{\mathrm{Eff}}(S)$ is polyhedral, then*

$$h^{1,1}(S) - \rho(S) = 2(p_g(S) + \gamma(X)).$$

If moreover $p_g(S) = 0$, by the Lefschetz $(1,1)$-theorem we have $\rho(S) = h^{1,1}(S)$. Thus by Corollary B we obtain

Theorem 7.14. *Let S be a product-quotient surface with $p_g(S) = 0$. If $\gamma(X) > 0$ then $\overline{\mathrm{Eff}}(S)$ is non-polyhedral, hence S is not a Mori dream space.*

In [8, Tables 2–5] one can find more than 100 families of surfaces with $p_g = q = 0$ and $\gamma > 0$. By Theorem 7.14, these surfaces are not Mori dream spaces. Moreover, none of them is of general type [8, Subsection 7.3].

To the best of our knowledge, only four examples of product-quotient surfaces of general type with $p_g = q = 0$ and $\gamma > 0$ are known to exist; they are described in Table 1. By Theorem 7.14, they are not Mori dream spaces.

Note that these examples are necessarily non-minimal (Remark 7.15). It is an interesting open problem to determine whether their minimal models are Mori dream spaces.

The last two surfaces, namely those with $\gamma(X) = 2$, contain precisely two (-1)-curves E_1 and E_2, where E_i is contracted by $f_i \colon S \longrightarrow C_i/G$. For each $i = 1, 2$, contracting E_i produces a surface S' with

$$p_g(S') = q(S') = 0, \quad K_{S'}^2 = 0,$$

which still satisfies the hypothesis of Corollary B.[7] Thus S' is not a Mori dream space either.

γ	K_S^2	$K_{\widetilde{S}}^2$	G	Sing(X)	t_1	t_2	Reference
1	1	3	$\mathrm{PSL}_2(\mathbb{F}_7)$	$1/7, 2/7, 4/7$	$3, 3, 7$	$2, 4, 7$	[7, Section 5]
1	0	1	$\mathrm{GL}_2(\mathbb{F}_4)$	$2 \times 1/6, 2 \times 1/2, 2/3$	$4, 4, 6$	$2, 4, 6$	[8, Section 7.1]
2	-1	1	$\mathbb{Z}_5 \times \mathbb{Z}_5$	$5 \times 1/5$	$5, 5, 5$	$5, 5, 5$	[8, Section 7.2]
2	-1	1	$\mathbb{Z}_5 \times \mathbb{Z}_5$	$5 \times 1/5$	$5, 5, 5$	$5, 5, 5$	[24, Theorem 5.5]

TABLE 1. Known examples of product-quotient surfaces of general type with $p_g = q = 0$ and $\gamma > 0$. Here $\widetilde{S}$ is the minimal model of S, whereas t_i denotes the signature of the G-cover $\varphi_i \colon C_i \longrightarrow C_i/G \simeq \mathbb{P}^1$. The fraction $\frac{a}{n}$ indicates a cyclic quotient singularity of type $\frac{1}{n}(1, a)$.

Remark 7.15. A product-quotient surface S of general type with $p_g(S) = q(S) = 0$ is minimal if and only if $\gamma(X) = 0$, see [8, Theorem 6.2]. These surfaces are listed in [7, Tables 1 and 2], and it is an interesting problem to classify those which are Mori dream spaces.

8. THE CRAIGHERO-GATTAZZO SURFACE

The Craighero-Gattazzo surface V is a particular numerical Godeaux surface, namely, a minimal surface of general type with

$$p_g(V) = q(V) = 0, \quad K_V^2 = 1.$$

[7] Let $f \colon S \longrightarrow C$ be a fibration and let $E \subset S$ be a (-1)-curve contracted by f. Then $f = f' \circ c$, where $c \colon S \longrightarrow S'$ is the contraction of E and f' is a fibration on S'. As $\rho(S) = \rho(S') + 1$ and $n(f) = n(f') + 1$, we see that $\rho(S) > n(f) + 2$ if and only if $\rho(S') > n(f') + 2$.

It was first constructed by Craighero and Gattazzo [18]. Later, it was studied by Dolgachev and Werner [21], who proved, among other things, that $H_1(V,\mathbb{Z}) = 0$ and claimed that V is simply connected. The proof of this last result contained a gap [22], and the fact that V is simply connected was proved only subsequently by Rana, Tevelev and Urzúa [47], using different methods.

The aim of this section is to show that the Craighero-Gattazzo surface is not a Mori dream space by applying Corollary B to a special genus 2 pencil on V, studied in [21]. We will use the notation of that paper throughout.

The Craighero–Gattazzo surface V is defined as the minimal resolution of singularities of a quintic surface $S \subset \mathbb{P}^3$ with four elliptic singularities; see [21, Section 2]. The quintic S is invariant with respect to the order 4 automorphism of $\mathbb{P}^3$

$$\sigma\colon [x:y:z:t] \mapsto [t:x:y:z]$$

which has the four fixed points

$$P_0 = [1:1:1:1], \quad P_1 = [1:i:-1:-i],$$
$$P_2 = [1:-i:-1:i], \quad Q_0 = [1:-1:1:-1].$$

The set of fixed points of

$$\sigma^2\colon [x:y:z:t] \mapsto [z:t:x:y]$$

is equal to the union of the two lines

$$r = \{x+z = y+t = 0\} = \langle P_1, P_2\rangle$$
$$r' = \{x-z = y-t = 0\} = \langle P_0, Q_0\rangle,$$

and the elliptic singularities of S are at the four reference points

$$a_1 = [1:0:0:0], \quad a_2 = [0:1:0:0], \quad a_3 = [0:0:1:0], \quad a_4 = [0:0:0:1].$$

Let

$$\pi\colon V \longrightarrow S$$

be the minimal resolution of singularities, and set $E_i := \pi^{-1}(a_i)$. The curves E_i are elliptic curves satisfying

$$E_i^2 = -1, \qquad K_V \cdot E_i = 1,$$

where the second equality follows from adjunction.

Let H denote a hyperplane section of S. Since S is a quintic hypersurface, adjunction gives $K_S = H$. Moreover, each a_i is a is a simple elliptic singularity of type $\widetilde{E}_8$; see [32, Theorem 7.6.4]. Since S is Gorenstein, we can write

$$K_V = \pi^* K_S + \sum_{i=1}^{4} \lambda_i E_i.$$

Intersecting with E_j and using $\pi^* K_S \cdot E_j = 0$, $K_V \cdot E_j = 1$, and $E_j^2 = -1$, we obtain $\lambda_j = -1$. Therefore

$$K_V = \pi^* H - E_1 - E_2 - E_3 - E_4.$$

The reader can find in [21, Section 2] a simple proof that V is of general type and satisfies $p_g(V) = q(V) = 0, K_V^2 = 1$. Moreover, we will later need the fact that V contains no (-2)-curves; see [21, Theorem 6.2] and [22]. Since V is a minimal surface of general type, this is equivalent to saying that K_V is ample.

The automorphisms σ and σ^2 of S both lift to automorphisms of V, which we denote by the same symbols. The automorphism $\sigma : V \longrightarrow V$ cyclically permutes the exceptional elliptic curves E_1, E_2, E_3, E_4, hence

$$\sigma^2(E_1) = E_3, \quad \sigma^2(E_2) = E_4. \tag{8.1}$$

The line r is contained in the quintic S and is disjoint from its singular locus. Therefore its inverse image

$$R := \pi^{-1}(r)$$

is a smooth rational curve. It is fixed pointwise by the involution $\sigma^2 : V \longrightarrow V$ and satisfies

$$K_V \cdot R = 1, \qquad R^2 = -3.$$

The line r' intersects S at five smooth points. Their inverse images on V are precisely the isolated fixed points of σ^2. We denote them by

$$\Sigma := \{Q_0, Q_1, Q_2, Q_3, Q_4\},$$

using the same notation for $Q_0 \in S$ and its inverse image on V.

Now we consider the complete linear system

$$|3K_V - R|$$

on V. It is a pencil without fixed part [21, Proposition 3.2]. Since $(3K_V - R)^2 = 0$, the pencil has no base points and therefore defines a fibration

$$f : V \longrightarrow \mathbb{P}^1.$$

By Bertini's theorem, its general fibre F is smooth, and the adjunction formula shows that it has genus 2, since

$$F^2 = 0, \quad K_V F = (3K_V - R)K_V = 2.$$

We are interested in the reducible fibres of f. As K_V is ample, a fibre F of $f : V \longrightarrow \mathbb{P}^1$ has at most $K_V F = 2$ irreducible components. In particular, if F is reducible, we have $F = A + A'$, with $K_V A = K_V A' = 1$. Moreover, since $(A + A')A = (A + A')A' = 0$, and $0 \leq p_a(A) = \frac{1}{2}(3 - AA')$, we get only two numerical possibilities:

(a) $AA' = -A^2 = -(A')^2 = 1$;
(b) $AA' = -A^2 = -(A')^2 = 3$.

Lemma 8.1. *Case* (b) *does not occur.*

Proof. We have seen that the fixed locus of the involution $\sigma^2 : V \longrightarrow V$ consists of the disjoint union of the smooth rational curve R and the five isolated points in

Σ. Then, using the topological fixed point formula [36, Theorem 1] and recalling that $q(V)=0$ implies $H^1(V,\mathbb{Q})=H^3(V,\mathbb{Q})=0$, we get

$$\begin{aligned}2+5=\chi_{\mathrm{top}}(\mathrm{Fix}(\sigma^2))&=\sum(-1)^q\,\mathrm{trace}\,((\sigma^2)^*\mid H^q(V,\mathbb{Q}))\\&=2+\mathrm{trace}\,((\sigma^2)^*\mid H^2(V,\mathbb{Q})),\end{aligned}$$

that is, $\mathrm{trace}\,((\sigma^2)^*\mid H^2(V,\mathbb{Q}))=5$.

On the other hand, since $p_g(V)=q(V)=0$ and $\chi_{\mathrm{top}}(V)=11$, we infer $b_2(V)=9$.

Thus, the invariant and anti-invariant subspaces $H^2(V,\mathbb{Q})^+$ and $H^2(V,\mathbb{Q})^-$ for the involution $(\sigma^2)^*\colon H^2(V,\mathbb{Q})\longrightarrow H^2(V,\mathbb{Q})$ have respective dimensions

$$\dim H^2(V,\mathbb{Q})^+=7,\quad \dim H^2(V,\mathbb{Q})^-=2.$$

By (8.1), a basis for $H^2(V,\mathbb{Q})^-$ consists of

$$\{E_1-E_3,\,E_2-E_4\}.$$

Suppose that there exists a reducible fibre $A+A'$ as in (b). Then

- $A+A'$ does not contain any of the five isolated fixed points of σ^2, as they lie on fibres of type (a) as shown in [21, Lemma 3.5, Corollary 3.6];[8]
- moreover, neither A nor A' equals R, since $(3K_V-R)R=6$. Since $A^2=(A')^2=-3$ is odd, [21, Lemma 3.4] implies that neither A nor A' is σ^2-invariant;
- $A+A'$ is σ^2-invariant, because every member of $|3K_V-R|$ is σ^2-invariant; see [21, p. 742].

Thus $A'=\sigma^2(A)$, that is $A-A'\in H^2(V,\mathbb{Q})^-$. Therefore there exist $\lambda,\mu\in\mathbb{Q}$ such that, in $H^2(V,\mathbb{Q})$:

$$A-A'=\lambda(E_1-E_3)+\mu(E_2-E_4).$$

Since $\lambda=(A-A')E_3$ and $\mu=(A-A')E_4$, both λ and μ are integers. Then

$$2(\lambda^2+\mu^2)=-(A-A')^2=-(A^2-2AA'+(A')^2)=12$$

provides the desired contradiction, because the equation $\lambda^2+\mu^2=6$ has no integral solutions. □

So, all reducible fibres of f are of type (a). We are now in a position to prove the main result of this section.

Theorem 8.2. *The pseudo-effective cone $\overline{\mathrm{Eff}}(V)$ of the Craighero-Gattazzo surface V is non-polyhedral. In particular, V is not a Mori dream space.*

Proof. All reducible fibres of f are of type (a), hence they are not 2-connected.

By the theory of genus-2 fibrations, the number of fibres of f that are not 2-connected is at most

$$K_V^2-2\chi(\mathscr{O}_V)+6=5,$$

[8] As K_V is ample, the cases (ii) and (iii) in [21, Lemma 3.5] do not occur; cf. [21, p. 751 and Remark 3.9].

see [14, Lemma 4.1(1) and Theorem 4.13]. Indeed, the divisor τ appearing there is supported on the points corresponding to the fibres that are not 2-connected, and its degree is $K_V^2 - 2\chi(\mathcal{O}_V) + 6$.

Since V contains no (-2)-curves, [21, Lemma 3.5 and Corollary 3.6] show that the five points of Σ lie on five distinct fibres, all of type (a). Therefore f has exactly five reducible fibres. Since each of them has two irreducible components, we obtain $n(f) = 5$.

On the other hand, since $p_g(V) = 0$, the Lefschetz (1, 1)-theorem gives

$$\rho(V) = h^{1,1}(V) = b_2(V) = 9.$$

Thus

$$\rho(V) = 9 > 7 = n(f) + 2,$$

and the result follows from Corollary B. □

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AI DISCLOSURE

All the ideas and the overall direction of this work were developed by the authors. During the preparation of the paper, the authors used ChatGPT (OpenAI) for technical assistance in checking calculations, identifying relevant references, and improving the exposition. All AI-assisted material was subsequently reviewed and edited by the authors, who independently verified every mathematical argument, calculation, and reference. The authors take full responsibility for the mathematical correctness and final text of the paper.

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